Ans: A

Q23

Note: A diode arranged in series with a resistor will give a half-wave rectified waveform with a rms current $I_{rms} = \frac{I_0}{2}$.

When the 1000 Ω resistor is conducting, peak current = 12/1000 = 0.012 A. Root mean square current is

$$I_1 = \frac{0.012}{2} = 0.0060 \text{ A (first half cycle)}$$

When the 2000 Ω resistor is conducting, peak current = 12/2000 = 0.0060 A. Root mean square current is

$$I_2 = \frac{0.0060}{2} = 0.0030 \text{ A (second half cycle)}$$

Total Average power $< P >= I_1^2 R_1 + I_2^2 R_2 = 0.006^2(1000) + 0.003^2(2000)$ = 0.054 W

Ans: B

Q24